matrixdiagrams.sty

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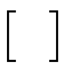
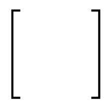
1 Main command

\matrixdiagram[scale][aspect]{content}

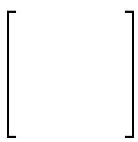
1.1 Scale (aspect = 1)



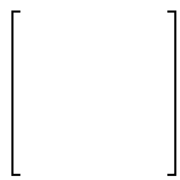
\matrixdiagram[1]{\mdBlank}



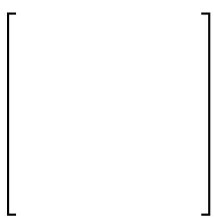
\matrixdiagram[2]{\mdBlank}



\matrixdiagram[3]{\mdBlank}



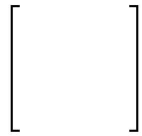
\matrixdiagram[4]{\mdBlank}



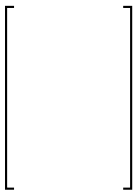
\matrixdiagram[5]{\mdBlank}

1.2 Aspect ratio (scale = 3)

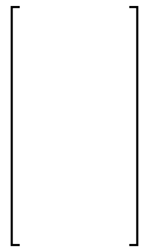
`\matrixdiagram[3][0.5]{\mdBlank}`



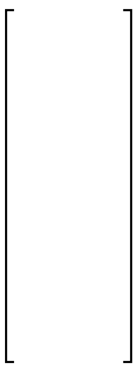
`\matrixdiagram[3][1]{\mdBlank}`



`\matrixdiagram[3][1.5]{\mdBlank}`



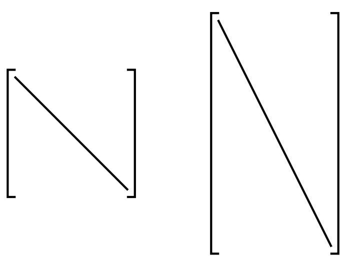
`\matrixdiagram[3][2]{\mdBlank}`



`\matrixdiagram[3][3]{\mdBlank}`

2 Diagonal commands

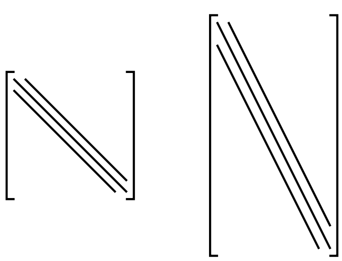
2.1 `\mdDiag` — main diagonal



`\matrixdiagram[3]{\mdDiag}`
`\matrixdiagram[3][2]{\mdDiag}`

2.2 `\mdDiagBand` — tridiagonal band

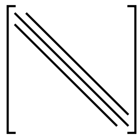
Three parallel anti-diagonals with spacing $d = 0.1$.



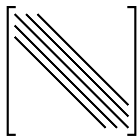
`\matrixdiagram[3]{\mdDiagBand}`
`\matrixdiagram[3][2]{\mdDiagBand}`

2.3 `\mdDiagBandN{n}` — variable band width

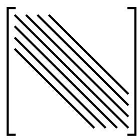
$n = 3, 5, 7, 9$ (spacing fixed at $d = 0.1$ per diagonal):



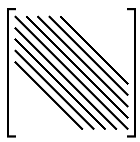
`\matrixdiagram[3]{\mdDiagBandN{3}}`



`\matrixdiagram[3]{\mdDiagBandN{5}}`

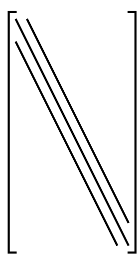


`\matrixdiagram[3]{\mdDiagBandN{7}}`

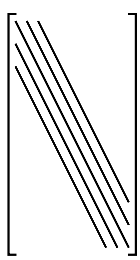


`\matrixdiagram[3]{\mdDiagBandN{9}}`

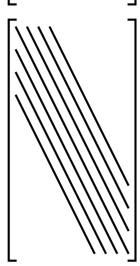
Note: `\mdDiagBandN{3}` is identical to `\mdDiagBand`. With aspect = 2:



`\matrixdiagram[3][2]{\mdDiagBandN{3}}`



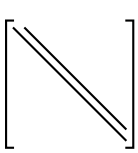
`\matrixdiagram[3][2]{\mdDiagBandN{5}}`



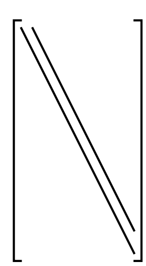
`\matrixdiagram[3][2]{\mdDiagBandN{7}}`

3 Bidiagonal commands

3.1 `\mdBidiagUpper` — main + super-diagonal

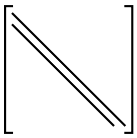


`\matrixdiagram[3]{\mdBidiagUpper}`

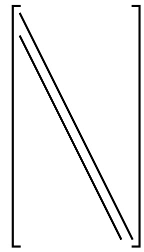


`\matrixdiagram[3][2]{\mdBidiagUpper}`

3.2 `\mdBidiagLower` — main + sub-diagonal



`\matrixdiagram[3]{\mdBidiagLower}`

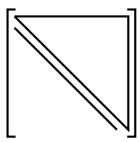


`\matrixdiagram[3][2]{\mdBidiagLower}`

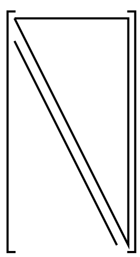
4 Hessenberg command

4.1 `\mdHessenberg` — outline only (default)

Upper-triangular outline $(0, H) \to (1, H) \to (1, 0)$ plus one sub-diagonal.

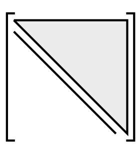


`\matrixdiagram[3]{\mdHessenberg}`

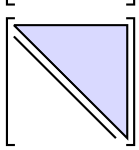


`\matrixdiagram[3][2]{\mdHessenberg}`

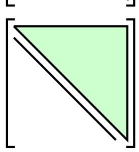
4.2 `\mdHessenberg[color]` — with fill



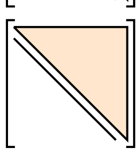
`\matrixdiagram[3]{\mdHessenberg[gray!15]}`



`\matrixdiagram[3]{\mdHessenberg[blue!15]}`



`\matrixdiagram[3]{\mdHessenberg[green!20]}`



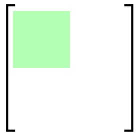
`\matrixdiagram[3]{\mdHessenberg[orange!20]}`

5 Fill and line commands

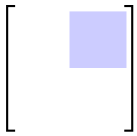
5.1 `\mdFill[color]{x0}{y0}{x1}{y1}`

Four quadrant fills; default color is `gray!20`:

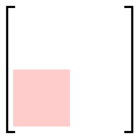
$y \in [0, 1]$.



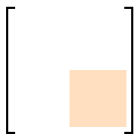
```
\matrixdiagram[3]{\mdFill[green!30]{0}{0.5}{0.5}{1}}
```



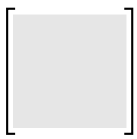
```
\matrixdiagram[3]{\mdFill[blue!20]{0.5}{0.5}{1}{1}}
```



```
\matrixdiagram[3]{\mdFill[red!20]{0}{0}{0.5}{0.5}}
```

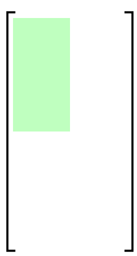


```
\matrixdiagram[3]{\mdFill[orange!25]{0.5}{0}{1}{0.5}}
```

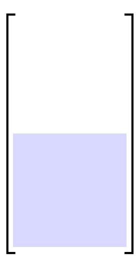


```
\matrixdiagram[3]{\mdFill{0}{0}{1}{1}}
```

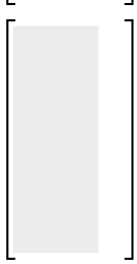
With aspect $= 2$ (y is still a fraction of the height, in $[0, 1]$):



```
\matrixdiagram[3][2]{\mdFill[green!25]{0}{0.5}{0.5}{1}}
```



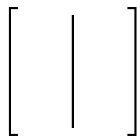
```
\matrixdiagram[3][2]{\mdFill[blue!15]{0}{0}{1}{0.5}}
```



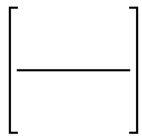
```
\matrixdiagram[3][2]{\mdFill[gray!15]{0}{0}{0.75}{1}}
```

5.2 `\mdVLine{x}` and `\mdHLine{y}`

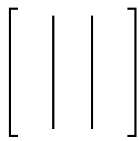
`\mdVLine{x}` runs the full height at $x \in [0, 1]$; `\mdHLine{y}` runs the full width at fractional height $y \in [0, 1]$.



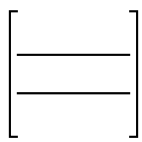
`\matrixdiagram[3]{\mdVLine{0.5}}`



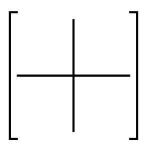
`\matrixdiagram[3]{\mdHLine{0.5}}`



`\matrixdiagram[3]{\mdVLine{0.33}\mdVLine{0.67}}`

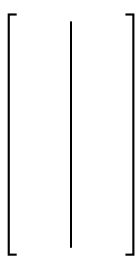


`\matrixdiagram[3]{\mdHLine{0.33}\mdHLine{0.67}}`

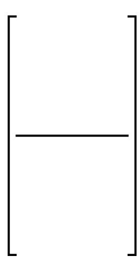


`\matrixdiagram[3]{\mdVLine{0.5}\mdHLine{0.5}}`

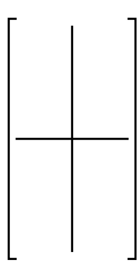
With aspect = 2:



`\matrixdiagram[3][2]{\mdVLine{0.5}}`

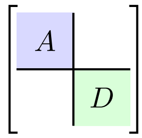


`\matrixdiagram[3][2]{\mdHLine{0.5}}`

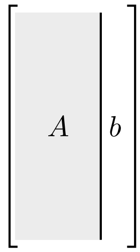


`\matrixdiagram[3][2]{\mdVLine{0.5}\mdHLine{0.5}}`

Combined with fills (block partitioning):



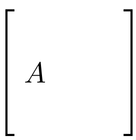

```
\matrixdiagram[3]{
  \mdFill[blue!15]{0}{0.5}{0.5}{1}
  \mdFill[green!15]{0.5}{0}{1}{0.5}
  \mdVLine{0.5}\mdHLine{0.5}
  \mdLabel{0.25}{0.75}{A}
  \mdLabel{0.75}{0.25}{D}
}
```



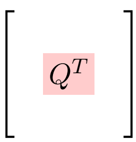

```
\matrixdiagram[3][2]{
  \mdFill[gray!15]{0}{0}{0.75}{1}
  \mdVLine{0.75}
  \mdLabel{0.375}{0.5}{A}
  \mdLabel{0.875}{0.5}{b}
}
```

6 Label commands

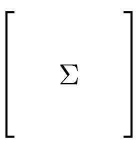
6.1 `\mdLabel{x}{y}{text}` — math mode



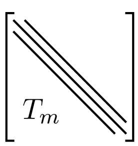

```
\matrixdiagram[3]{\mdLabel{0.2}{0.5}{A}}
```




```
\matrixdiagram[3]{\mdLabel[red!20]{0.5}{0.5}{Q^T}}
```




```
\matrixdiagram[3]{\mdLabel{0.5}{0.5}{\Sigma}}
```




```
\matrixdiagram[3]{\mdDiagBand\mdLabel{0.25}{0.2}{T_m}}
```

6.2 `\mdLabelCenter{text}` — always centered

Automatically placed at (0.5, $H/2$) regardless of aspect ratio:

$\begin{bmatrix} Q \end{bmatrix}$ `\matrixdiagram[3][1]{\mdLabelCenter{Q}}`

$\begin{bmatrix} \\ Q \\ \\ \end{bmatrix}$ `\matrixdiagram[3][2]{\mdLabelCenter{Q}}`

$\begin{bmatrix} \\ \\ Q \\ \\ \\ \end{bmatrix}$ `\matrixdiagram[3][3]{\mdLabelCenter{Q}}`

$\begin{bmatrix} Q \end{bmatrix}$ `\matrixdiagram[3][0.5]{\mdLabelCenter{Q}}`

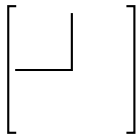
6.3 `\mdLabelRaw{x}{y}{text}` — raw text

$\begin{bmatrix} \text{dense} \end{bmatrix}$ `\matrixdiagram[3]{\mdLabelRaw{0.5}{0.5}{dense}}`

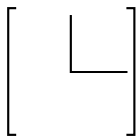
$\begin{bmatrix} \approx 0 \end{bmatrix}$ `\matrixdiagram[3]{\mdLabelRaw{0.5}{0.5}{$\approx 0$}}`

7 Spike commands

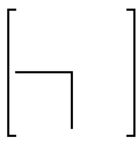
`\mdSpike{x}{y}{dir}` draws an L-shaped corner at (x, y). The four directions connect to different pairs of edges:



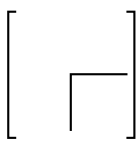
`\matrixdiagram[3]{\mdSpike{0.5}{0.5}{tl}}`



`\matrixdiagram[3]{\mdSpike{0.5}{0.5}{tr}}`

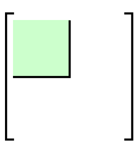


`\matrixdiagram[3]{\mdSpike{0.5}{0.5}{bl}}`

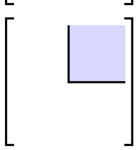


`\matrixdiagram[3]{\mdSpike{0.5}{0.5}{br}}`

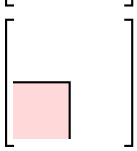
`tl` (top+left), `tr` (top+right), `bl` (bottom+left), `br` (bottom+right).
With fills:



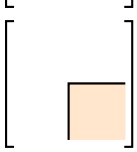
`\matrixdiagram[3]{\mdFill[green!20]{0}{0.5}{0.5}{1}\mdSpike{0.5}{0.5}{tl}}`



`\matrixdiagram[3]{\mdFill[blue!15]{0.5}{0.5}{1}{1}\mdSpike{0.5}{0.5}{tr}}`

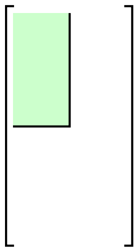


`\matrixdiagram[3]{\mdFill[red!15]{0}{0}{0.5}{0.5}\mdSpike{0.5}{0.5}{bl}}`

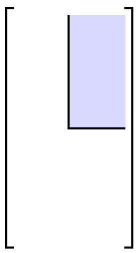


`\matrixdiagram[3]{\mdFill[orange!20]{0.5}{0}{1}{0.5}\mdSpike{0.5}{0.5}{br}}`

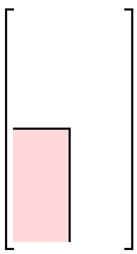
With aspect = 2, corner at mid-height $(0.5, 0.5)$:



```
\matrixdiagram[3][2]{\mdFill[green!20]{0}{0.5}{0.5}{1}\mdSpike{0.5}{0.5}{tl}}
```



```
\matrixdiagram[3][2]{\mdFill[blue!15]{0.5}{0.5}{1}{1}\mdSpike{0.5}{0.5}{tr}}
```

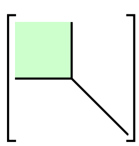


```
\matrixdiagram[3][2]{\mdFill[red!15]{0}{0}{0.5}{0.5}\mdSpike{0.5}{0.5}{bl}}
```

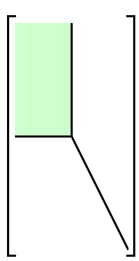
8 Clipped diagonal commands

The `...From` variants force the diagonal band from $[x_1, y_1]$ to $[x_2, y_2]$. Always pair with `\mdSpike` at the same corner.

8.1 `\mdDiagFrom{x1}{y1}{x2}{y2}`

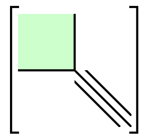


```
\matrixdiagram[3]{
  \mdFill[green!20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{tl}
  \mdDiagFrom{0.5}{0.5}{1}{0}
}
```

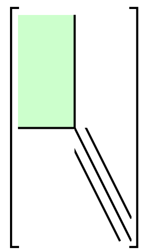


```
\matrixdiagram[3][2]{
  \mdFill[green!20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{tl}
  \mdDiagFrom{0.5}{0.5}{1}{0}
}
```

8.2 \mdDiagBandFrom{cx}{cy}



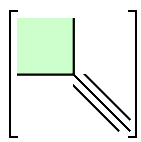
```
\matrixdiagram[3]{
  \mdFill[green!20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{tl}
  \mdDiagBandFrom{0.5}{0.5}{1}{0}
}
```



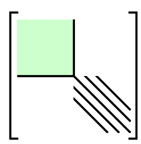
```
\matrixdiagram[3][2]{
  \mdFill[green!20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{tl}
  \mdDiagBandFrom{0.5}{0.5}{1}{0}
}
```

8.3 \mdDiagBandNFrom{n}{cx}{cy}

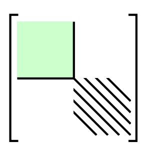
$n = 3,\ 5,\ 7$:



```
\matrixdiagram[3]{
  \mdFill[green!20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{tl}
  \mdDiagBandNFrom{3}{0.5}{0.5}{1}{0}
}
```



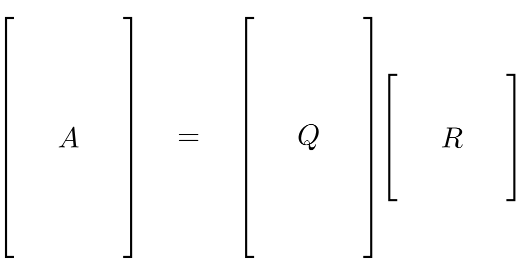
```
\matrixdiagram[3]{
  \mdFill[green!20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{tl}
  \mdDiagBandNFrom{5}{0.5}{0.5}{1}{0}
}
```



```
\matrixdiagram[3]{
  \mdFill[green!20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{tl}
  \mdDiagBandNFrom{7}{0.5}{0.5}{1}{0}
}
```

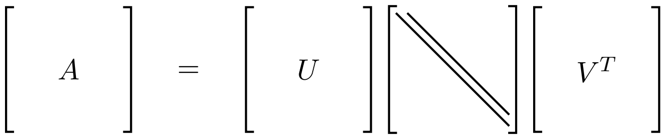
9 Realistic equations

9.1 QR decomposition: $A = QR$



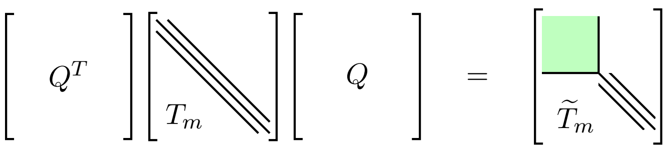

```
\matrixdiagram[3][2]{\mdLabelCenter{A}} =
\matrixdiagram[3][2]{\mdLabelCenter{Q}}
\matrixdiagram[3]{\mdLabelCenter{R}}
```

9.2 Bidiagonalization: $A = UBV^T$



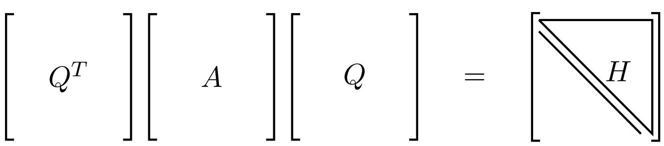
```
\matrixdiagram[3]{\mdLabelCenter{A}} =
\matrixdiagram[3]{\mdLabelCenter{U}}
\matrixdiagram[3]{\mdBidiagUpper}
\matrixdiagram[3]{\mdLabelCenter{V^T}}
```

9.3 Similarity: $Q^T T_m Q = \widetilde{T}_m$



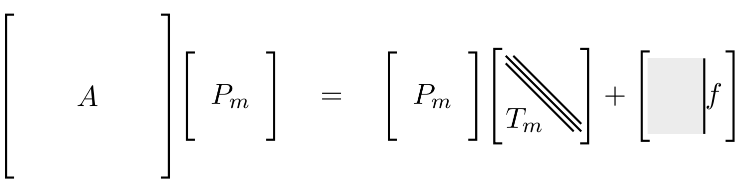
```
\matrixdiagram[3]{\mdLabelCenter{Q^T}}
\matrixdiagram[3]{
  \mdDiagBand\mdLabel{0.25}{0.15}{T_m}
}
\matrixdiagram[3]{\mdLabelCenter{Q}} =
\matrixdiagram[3]{
  \mdFill[green!25]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{tl}
  \mdDiagBandFrom{0.5}{0.5}{1}{0}
  \mdLabel{0.3}{0.12}{\widetilde{T}_m}
}
```

9.4 Hessenberg reduction: $Q^T AQ = H$



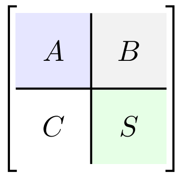
```
\matrixdiagram[3]{\mdLabelCenter{Q^T}}
\matrixdiagram[3]{\mdLabelCenter{A}}
\matrixdiagram[3]{\mdLabelCenter{Q}} =
\matrixdiagram[3]{
  \mdHessenberg\mdLabel{0.7}{0.55}{H}
}
```

9.5 Column-bordered decomposition: $AP_m = P_m T_m + f$



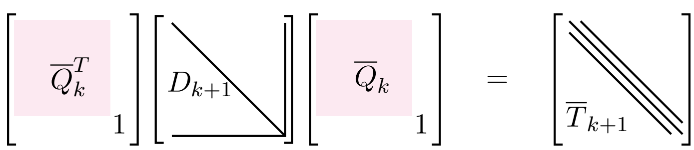
```
\matrixdiagram[4]{\mdLabelCenter{A}}
\matrixdiagram[2]{\mdLabelCenter{P_m}} =
\matrixdiagram[2]{\mdLabelCenter{P_m}}
\matrixdiagram[2]{
  \mdDiagBand \mdLabel{0.25}{0.15}{T_m}
} +
\matrixdiagram[2]{
  \mdFill[gray!15]{0}{0}{0.75}{1}
  \mdVLine{0.75} \mdLabel{0.875}{0.5}{f}
}
```

9.6 Schur complement structure




```
\matrixdiagram[4]{
\mdFill[blue!10]{0}{0.5}{0.5}{1} \mdFill[green!10]{0.5}{0}{1}{0.5}
\mdFill[gray!10]{0.5}{0.5}{1}{1}
\mdVLine{0.5} \mdHLine{0.5}
\mdLabel{0.25}{0.75}{A} \mdLabel{0.75}{0.75}{B}
\mdLabel{0.25}{0.25}{C} \mdLabel{0.75}{0.25}{S}}
```

9.7 Bordered similarity: $\overline{Q}_k^T D_{k+1} \overline{Q}_k = \overline{T}_{k+1}$




```
\matrixdiagram[3]{
  \mdFill[magenta!10]{0}{0.15}{0.85}{1}
  \mdLabel{0.5}{0.5}{\overline{Q}_{k}^T}
  \mdLabel{0.93}{0.08}{1}
}
\matrixdiagram[3]{
  \mdDiag
  \draw[thick] (0,0) -- (1,0) -- (1,\mdH);
  \mdLabel{0.25}{0.45}{D_{k+1}}
}
\matrixdiagram[3]{
  \mdFill[magenta!10]{0}{0.15}{0.85}{1}
  \mdLabel{0.5}{0.5}{\overline{Q}_{k}}
  \mdLabel{0.93}{0.08}{1}
} =
\matrixdiagram[3]{
  \mdDiagBand
  \mdLabel{0.25}{0.15}{\overline{T}_{k+1}}
}
```

10 Command reference

`\matrixdiagram[s][a]{c}`	Main command. `s` = scale (def. 1), `a` = aspect (def. 1 = square). Position args below are fractions $x, y \in [0, 1]$, any aspect.
`\mdDiag`	Anti-diagonal $(0, H) \to (1, 0)$.
`\mdDiagBand`	Tridiagonal band, spacing $d = 0.1$.
`\mdDiagBandN{n}`	n-diagonal band (n odd), fixed step 0.1.
`\mdBidiagUpper`	Main diagonal + super-diagonal ($d = 0.1$).
`\mdBidiagLower`	Main diagonal + sub-diagonal ($d = 0.1$).
`\mdHessenberg`	Upper-triangular outline + sub-diagonal.
`\mdHessenberg[color]`	Same with filled interior.
`\mdDiagFrom{x1}{y1}{x2}{y2}`	Main diagonal, from (x_1, y_1) to (x_2, y_2)
`\mdDiagBandFrom{x1}{y1}{x2}{y2}`	Tridiagonal band, clipped.
`\mdDiagBandNFrom{n}{x1}{y1}{x2}{y2}`	n-diagonal band, clipped.
`\mdFill[col]{x0}{y0}{x1}{y1}`	Filled rectangle. Default color `gray!20`.
`\mdVLine{x}`	Vertical line, full height.
`\mdHLine{y}`	Horizontal line, full width, at fractional height y.
`\mdLabel{x}{y}{text}`	Math-mode label at (x, y).
`\mdLabel[color]{x}{y}{text}`	Same with filled node label
`\mdLabelRaw{x}{y}{text}`	Raw-text label at (x, y).
`\mdLabelCenter{text}`	Math label at center (0.5, $H/2$), any aspect.
`\mdSpike{x}{y}{dir}`	L-corner; `dir` $\in$ {`tl, tr, bl, br`}.
`\mdBlank`	Empty content (brackets only).
`\mdH`	Expands to current aspect value inside content.